

Control plane procedures run sequentially by design and the data plane packet delivery cannot start until these procedures are completed. The paper proposes DPCM, which aims to reduce data access latency through parallel processing and exploiting device side state replica. The author argues that the control plane operations are vital in providing data access in mobile networks. For each device on the network, the control plane establishes sessions which install states for all involved devices and nodes.

Control plane operations are inherently slow because are mandated to run in sequential order, which is inefficient for three reasons. Firstly, some control plane procedures are not mandatory for starting or resuming data service but they have to be completed before data delivery starts in current design of mobile networks. Secondly, some procedures can run in parallel but are mandated to run sequentially. Thirdly, in case of failures, the entire control sequence is blocked, increasing latency. The key observation of the authors is that the data plane forwarding can start without waiting for all control procedures.

The authors present DPCM which treats these problems as a state management issue and accelerates data access through three main techniques. First is to bypass some control plane procedures and substitute them with equivalent, faster operations. For this purpose, DPCM retrieves the state replica already available on the device side early and replicates it on nodes of interest. Second technique is to pipeline some control plane signalling procedures with data plane packet delivery. Thirdly, DPCM parallelizes the mandatory control plane procedures. DPCM is implemented as an extension to Open Air Interface, a software defined cellular protocol stack, and achieves 2.1x to 11.5x latency reduction.

Mobile networks experience three types of latency; Service Establishment (C1), Handover(C2) and Wide Area Roaming (C3). C1 is initiated per data session. Not all procedures in this step are necessary, existing state replica on device can be used to accelerate these procedures. C2 is triggered by mobility. Ideally, planned handovers between base stations should incur very little overhead but due to transient radio outages, these overheads are increased. State replica on device can also be leveraged here. In C3, user's location update requires execution of a sequence of control plane procedures to resume data service. Sequential state copy between mobility controllers can be parallelized to reduce latency in this case.

DPCM leverages state replicas available on device and other nodes, and parallelizes control plane procedures to reduce mobile network latency. DPCM guarantees worst case latency (in case of conflicts) to be no worse than LTE. Experimentation results show that DPCM reduces average web page loading time by up to 3.7x and videos by up to 2.1x. DPCM has minimal memory and CPU overhead and negligible power consumption as it reuses 4G's existing communication mechanisms.

The techniques discussed in the paper appear analogous to dynamic programming and it is interesting to see such impactful application of the concept. Mobile internet is the primary mode of internet access for many and improving on its design while leveraging the existing infrastructure can have a huge impact for a lot of people.